

QUOTES RECOMMENDATION CHATBOT USING NLP

Epic 3: DATA COLLECTION & MODEL BUILDING

Story 1: Data Collection – Creating User Queries (nlu.yml)

1. Introduction

Data collection is the first and most important step in training the chatbot. Since the Quotes Recommendation Chatbot is designed to provide quotes based on user emotions and preferences, it must understand different ways users may express their needs.

Users may request quotes in various forms such as motivation, inspiration, love, humor, or success. Therefore, collecting diverse user query examples is essential for accurate intent classification.

2. Understanding the nlu.yml File

In Rasa, user training data is stored in the nlu.yml file using YAML format.

This file contains:

- Intents (purpose of user message)
- Example sentences for each intent
- Variations of how users might ask for the same thing

The chatbot learns patterns from these examples and uses them to classify future user inputs correctly.

3. Intents Defined for the Project

The following intents are created for the Quotes Recommendation Chatbot:

- greet – User greets the chatbot
- motivation – User requests motivational quotes
- inspiration – User requests inspirational quotes
- love – User requests love-related quotes
- funny – User requests humorous quotes
- success – User requests success-related quotes

- goodbye – User ends the conversation

Each intent includes multiple example sentences to improve recognition accuracy.

4. Sample nlu.yml Structure

Below is a sample structure used in the project:

version: "3.1"

nlu:

- intent: greet

examples: |

- hello
- hi
- hey there
- good morning
- good evening

- intent: motivation

examples: |

- I need motivation
- Give me a motivational quote
- I feel demotivated
- Motivate me
- Something inspiring for work

- intent: inspiration

examples: |

- Inspire me
- Give me an inspirational quote
- I need some inspiration
- Share something uplifting

- intent: funny

examples: |

- Tell me something funny
- I want a humorous quote
- Make me laugh
- Say something funny

- intent: success

examples: |

- Give me a success quote
- I want something about success
- Inspire me to achieve success
- Share a success thought

- intent: goodbye

examples: |

- bye
- goodbye
- see you later
- talk to you later

5. Importance of This Step

This step is crucial because:

- It directly impacts intent recognition accuracy
- It improves chatbot understanding of natural language variations
- It reduces misclassification errors
- It enhances user experience

Providing diverse and realistic examples ensures that the chatbot can handle different user expressions effectively.

6. Conclusion

Creating the nlu.yml file with well-defined intents and multiple example queries forms the foundation of the chatbot's intelligence.

Accurate and diverse training data enables the Quotes Recommendation Chatbot to correctly understand user needs and provide meaningful, context-aware quote recommendations.